

RENANA FRIEDMAN

Backend Developer

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SUMMARY

Backend Developer with a BSc in Software Engineering (GPA 90), hands-on experience shipping production systems, and a self-built AI-powered RAG application. Fast learner who takes full ownership and works well within a team.

PROJECT

AI Portfolio Assistant - RAG + Real-Time Avatar | 2026
[Live site](#) | [GitHub - Backend](#) | [GitHub - Frontend](#)

A RAG-powered chat assistant embedded in my portfolio site - visitor questions are answered by an LLM using retrieved context, then streamed live through a speaking video avatar.

- ⇒ Built a FastAPI and LangChain RAG pipeline (PostgreSQL/pgvector, OpenAI gpt-4o-mini) for context-aware answers about my background.
- ⇒ Streamed real-time avatar responses via D-ID's WebRTC API (SDP/ICE), avoiding the latency of file-based video generation.
- ⇒ Built the React (Vite) frontend and deployed the full stack (API, frontend, database) on Railway.

EXPERIENCE

Backend Developer | SmartyApp 2025 - 2026

- ⇒ Built a WhatsApp bot backend from scratch in TypeScript using Meta's official Cloud API, including persistent conversation state and media handling (image upload/retrieval).
- ⇒ Integrated the bot with the company's e-commerce site and AWS S3 for image storage and retrieval.
- ⇒ Owned architectural decisions end-to-end, translating client requirements into implementation.

SKILLS

- ⇒ **Languages:** Java, Python, JavaScript, TypeScript, C#
- ⇒ **Backend & APIs:** Spring Boot, FastAPI, REST APIs, Node.js
- ⇒ **AI / LLM:** LangChain, OpenAI API, Retrieval-Augmented Generation (RAG), Vector Search
- ⇒ **Databases:** PostgreSQL, pgvector, MySQL
- ⇒ **DevOps & Tools:** Docker, Git, Swagger, Kafka, Railway (deployment), Postman
- ⇒ **Frontend:** React, Vite, CSS

EDUCATION

Backend Development Course | 2025

- ⇒ Built REST APIs, managed databases, and learned scalable architecture design.
- ⇒ Hands-on work with Spring Boot, Docker, Swagger, Kafka, and Python.

BSc Software Engineering | Jerusalem College of Technology 2020 - 2023

- ⇒ GPA: 90
- ⇒ Selected Courses: Algorithm Analysis & Complexities, Data Mining, Big Data, Neural Networks and Deep Learning, Introduction to AI, Software Engineering, Full-Stack Web Development.

ADDITIONAL PROJECTS

Allergen Identification through Image Processing | [GitHub](#) / 2022

- ⇒ Developed a mobile app that detects allergens in food products by analyzing ingredient lists from photos.

Management System for a Bus Company | [GitHub](#) / 2021

- ⇒ Built a layered WPF desktop application for managing a public bus network - buses, lines, stations, and scheduling - with clean PL/BL/DAL architecture and Singleton, Factory, and Adapter design patterns.

LANGUAGES

- ⇒ Hebrew - Native
- ⇒ English - Professional